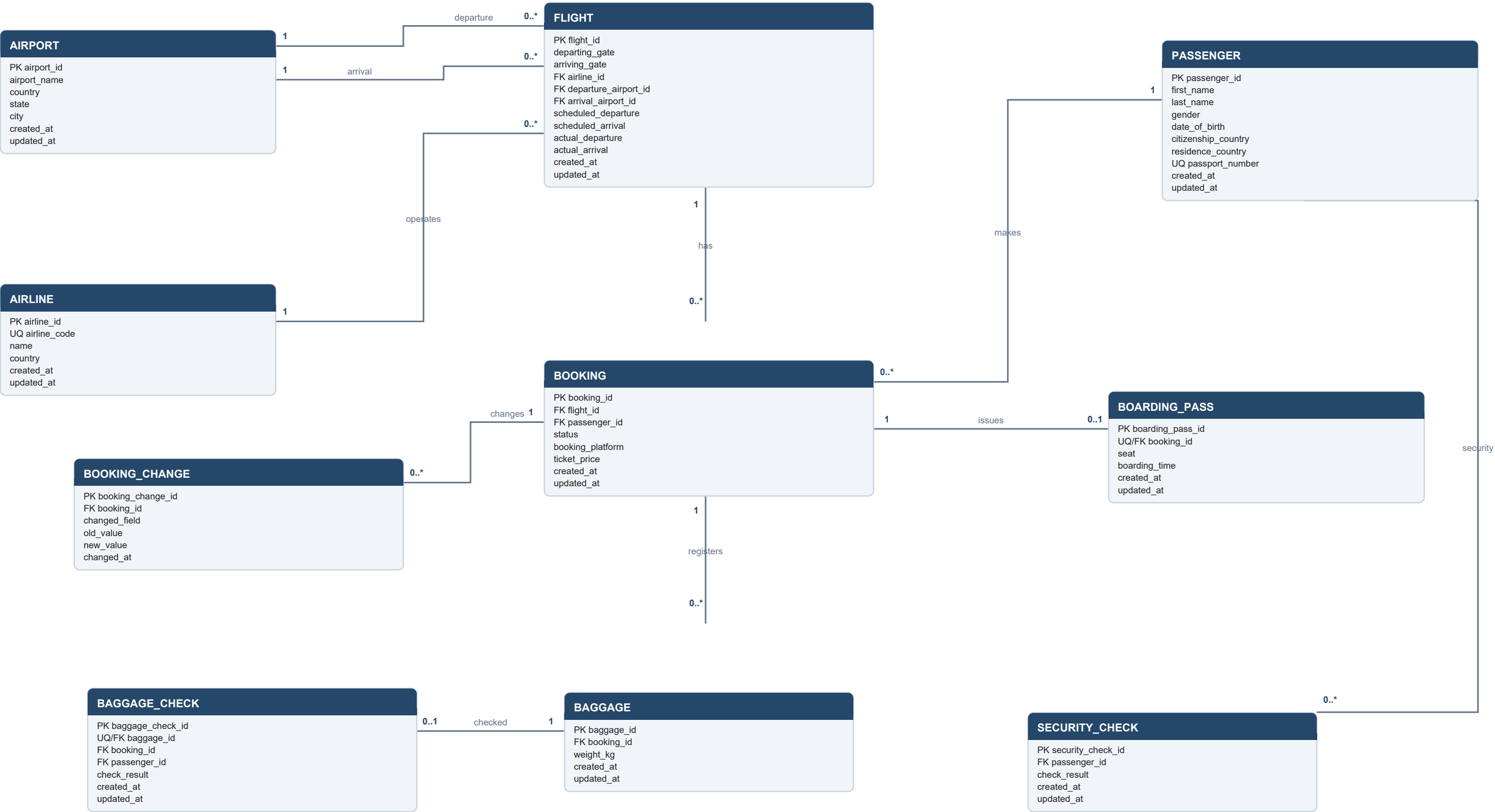


Laboratory Work 1

ER Diagram for an International Airport Database

Student: Мәди Мурат

ER Diagram of the Airport Database



PK - primary key, FK - foreign key, UQ - unique. 0..* means zero or many; 0..1 means zero or one.

3 Entities and attributes

The table below shows the main attributes and keys. Common data types are BIGINT for IDs, VARCHAR for text, DATE for birth date, TIMESTAMP for date and time, and NUMERIC for price and baggage weight.

Entity	Main attributes	Keys
Airport	airport_name, country, state, city, created_at, updated_at	PK airport_id
Airline	airline_code, name, country, created_at, updated_at	PK airline_id; UQ airline_code
Flight	gates, scheduled and actual departure/arrival times, timestamps	PK flight_id; FK airline_id and two airports
Passenger	name, gender, date of birth, citizenship, residence, passport	PK passenger_id; UQ passport_number
Booking	status, booking_platform, ticket_price, timestamps	PK booking_id; FK flight_id, passenger_id
BookingChange	changed_field, old_value, new_value, changed_at	PK booking_change_id; FK booking_id
BoardingPass	seat, boarding_time, timestamps	PK boarding_pass_id; FK booking_id
Baggage	weight_kg, timestamps	PK baggage_id; FK booking_id
BaggageCheck	check_result, timestamps	PK baggage_check_id; FK baggage_id, booking_id, passenger_id
SecurityCheck	check_result, timestamps	PK security_check_id; FK passenger_id

4 Relationships

Relationship	Type	Explanation
Airline - Flight	1 : M	One airline can operate many flights.
Airport - Flight	1 : M	An airport can be used by many flights for departure or arrival.
Flight - Booking	1 : M	One flight can have many bookings.
Passenger - Booking	1 : M	One passenger can make many bookings.
Booking - BookingChange	1 : M	A booking can have several change records.
Booking - BoardingPass	1 : 0..1	A booking may have one boarding pass.
Booking - Baggage	1 : M	A booking may contain several baggage items.
Baggage - BaggageCheck	1 : 0..1	A baggage item may have one check result.
Passenger - SecurityCheck	1 : M	A passenger may pass security more than once.

5 Main constraints

Primary keys must be unique. Foreign keys must refer to existing records. Passport number and airline code are unique. Ticket price cannot be negative, and baggage weight must be greater than zero. The arrival airport must be different from the departure airport.

6 Conclusion

In this work I created an ER diagram for an international airport database. I identified the main entities, added their attributes, selected primary and foreign keys, and showed the relationships between the tables. This model can be used as a basis for creating the database in a DBMS.